

Computational Policy Process Studies

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April 2021

8100 words

Abstract

While policy process theory has converged on the view that policymaking can be studied as a complex system, the literature has only minimally used the methodological complement to the theory - experiments performed with computational models. Implementations are rare, mainly pushed by computer scientists in trans-disciplinary work and often so detached from mainstream theory that they form a separate line of research instead of testing theories from the social sciences. This paper builds on the theory of policy processes and computational sciences to advance the computational turn of policy process studies. We examine how and why complexity science lends itself to study policymaking, propose a workflow to guide the creation of computational policy process models, describe the contours of a computational approach to policy process modeling and define goals for future research that follow from this computational turn. Overall, we aim to promote a computational turn of policy process studies that is empirical and hypothesis-driven.

Introduction

An important development has marked peace and conflict studies, policy analysis and migration studies in recent years: the use of computational methods to collect, analyse and generate data to explore social phenomena, often summarized as ‘computational social science’ CSS (Conte et al., 2012; Lazer et al., 2020). This paper contributes to such a computational turn in policy process studies. Policy process studies (PPS) refers to the scholarship that examines the cognitive, behavioral and social processes that underpin public policymaking within governments and international organizations (Howlett, 2020). While often neglected at the benefit of policy analysis (e.g. the evaluation of policies’ effectiveness) (Cairney & Weible, 2017), PPS sheds light on the determinants of (un)successful collective action.

In the 21st century alone, humanity has witnessed a vast range of challenges: a global economic crisis, a global pandemic, accelerating climate change and accelerating technological progress. To make sure technology is beneficial and shocks can be coped with, it is essential to better understand how societies can achieve the governance of such transgenerational global public goods (Bostrom, 2013).

The computational turn that we articulate in this paper finds its anchoring in the convergence of policy process theory over the past decade. Explicitly or implicitly, various theories of the policy process unite in the view that policy processes satisfy the hallmarks of complex systems (Cairney, 2012; Geyer & Rihani, 2010; Morçöl, 2013; Teisman & Klijn, 2008). Complex systems consist of many heterogeneous parts whose interactions in networks lead to the emergence of macro-level outcomes, often in a nonlinear fashion (Geyer & Rihani, 2010; B. D. Jones & Baumgartner, 2005b). In the realm of policymaking, this means individuals and groups interact dynamically to create policies.

Complex systems can be described by complexity theory which has its origins in physics before being successfully applied to chemical, biological (Mitchell, 2011) and social systems (Helbing et al., 2015), e.g., in the study of conflict (c.f. Stauffer, 2021). Looking at the erratic pattern of public budgets (B. D. Jones et al., 2009) and the size and complexity of policy networks (Koliba, Meek, Zia, & Mills, 2018; Van Waarden, 1992), it becomes apparent that complexity theory is an appropriate lens for policy process research.

With applications of complexity theory usually come computational methods that enable the formalization of system mechanisms and, through algorithmic processes, the generation of emergent patterns at the system level (Epstein, 2006; Page, 2018). Together, complexity theory and computational methods form the field of complexity science which aims to identify laws and systemic signatures across complex systems (Thurner, Hanel, & Klimek, 2018). Even though the theory of complexity has been extensively applied to policy process studies, the field lacks computational applications. Only then will the actual science of complexity be used to study collective action. The purpose of this paper is to facilitate this transition.

Facilitating the application of complexity science to policy process studies promises exciting scientific progress by furthering Herbert Simon’s vision of policy process studies. Simon contributed pioneering work on the fundamentals of bounded rationality (Simon, 1957), its applications to policymaking (Simon & March, 1976), the architecture of complex systems (Simon, 1991), the mechanism of preferential attachment in networks leading to power-law distributions (Simon, 1955), and computational methods (Simon, 1969). In this, Simon’s view was hypothesis-driven, based on his empirical observations and relied on the formalization of complex systems and the study of their unpredictability (B. D. Jones, 2003; Mintrom, 2015; Moynihan, 2018). With the exception of a handful of scientists (e.g. B. D. Jones and Baumgartner (2005b); Rhodes and Marsh (1992); Thomas (2017)),

scholarship in policy process studies has delved instead more into critical, narrative and qualitative analyses (Beardsworth, 2020) or purely rational, simplistic approaches (Enserink, Koppenjan, & Mayer, 2013), thus moving away from Simon’s approach to social sciences. We here advocate to re-engage with the early work of one of the fathers of the field to help enrich the study of policy processes moving forward.

In other domains, such as conflict and migration, the computational turn is developing in parallel to the mainstream (Lazer et al., 2020). Mathematicians, computer scientists and physicists are attracted by the challenge of understanding social phenomena, the potential social impact thereof, and the new challenge of applying their methods to social systems (Watts, 2016). While this trans-disciplinary interest is laudable, it often is highly detached from decades of literature tackling the same questions, albeit in a less technical manner. In this paper, we argue for a computational turn in its truest sense, i.e. building upon the iterative theory-building that has characterised policy process studies for almost a century. It includes the work by, for instance, Sabatier and Weible (2018), Jones & Baumgartner (2009), Geyer & Rihani (2010), Morçöl (2013), Cairney (2012), Geyer & Cairney (2015), and Teisman & Klijn (2008). In recent literature, authors have started to portray this computational turn and made theoretical strides in shaping it in the most valuable way possible (Johnson, 2015). Our paper builds on them to pave the way for this advancement.

A computational turn of policy process studies suggests either a change in the education of social scientists (e.g. by providing computational method courses to undergraduate and graduate students) or a change in the formation of research teams. The authors of this paper are bringing together their backgrounds in political science, behavioral science, mathematics, physics, computer science and machine learning, exemplifying such teams. We hope that this paper speaks to the reality, interests and sensitivities of the different fields that need to contribute to achieve this advancement of policy process studies.

This paper is divided into four sections. First, we survey the literature that portrays policy processes as complex systems in examining complexity, their added-value and shortcomings. Second, we provide a short introduction to computational models and why they are particularly well-suited to study social systems. Third, we outline the contours of a computational approach by linking examples of computational methods to phenomena of common interest in the policy process literature. Fourth, we sketch an agenda for further research, including general guidance for the research process as well as key research questions.

1 Complexity in policy processes and methodological frontiers

Policy processes consist of the interplay between the various forces that characterise policymaking systems. They notably include a vast variety of policy actors and their surrounding environment that nudges or incentives them to behave in a particular way and make certain choices. Similarly to conflicts, these dynamics have aroused interest among social scientists because they are characteristic of heterogeneous individuals aligning to take collective action. Where conflicts are characterized by individuals who may outright kill each other, the stakes of policymaking are less salient. The process of figuring out how best to use pooled resources is vital to understand, as they define the visions of our constitutions, societal narratives, common goals, specify the instruments for collective action and whether their outputs are beneficial to society (Bednar, 2012; Gintis, Doebeli, & Flack, 2012).

Examining policymaking processes may also increase legitimacy by fostering transparency and

spotting where they may be broken (Cairney, 2019; Dye, 1992). As policy processes often involve conflicts of subjective value judgements, objective analyses of underlying processes can pin-point reasons for disagreements, and subsequently reduce or outright collusion. Scholars of policy processes have advanced many theories, case studies and empirical analyses to describe such convoluted phenomena (Weible & Sabatier, 2018). In this section, we survey why those processes satisfy the hallmarks of complex systems and sketch current methodological frontiers in addressing the complexity of policymaking.

1.1 Theoretical convergence: policy processes as complex systems

Theories of policy processes converge on their conceptualization as complex systems. They, however, do not always do so explicitly (Cairney, 2012). That is, they sometimes propose other frameworks that broadly rely on the same ideas but without using the language of complex systems. In what follows, we first introduce the hallmarks of complex systems before describing how it is used - often implicitly - in policy processes studies (summarised in Table 2).

Complex systems satisfy two key properties: (1) many, heterogeneous, interacting and adaptive parts that lead to (2) the emergence of system-level dynamics and properties (Mitchell, 2011; Pines, 2018; Thurner et al., 2018). Explanations of such systems need to be *reduced* by identifying the key mechanisms that govern such dynamics, instead of isolating each moving parts or system levels (e.g. micro versus macro). As such, complexity science can describe otherwise inexplicable phenomena in physical, chemical, biological and social systems.

First, complex systems have many constituent parts. In chemical systems, such parts are molecules while in social systems they are individuals or groups. These parts can be heterogeneous; have sets of characteristics that vary from part to part. In chemical systems, for example, characteristics of molecules are atomic composition and ionization while, in social systems, characteristics of individuals include beliefs, interests, roles and more. Those components interact with one another. In chemical systems, for example, molecules of hydrogen and oxygen interact in a glass of water. In social systems, individuals may interact spatially (e.g. neighbours) or through network structures. Lastly, such parts adapt their behavior as a function of their interactions and environment. In chemical systems, molecules exchange atoms and electrons while, in social systems, individuals may change their mind or their behavior as a function of social cues. The described processes characterise the micro-level dynamics of complex systems.

Second, these micro-level dynamics lead to the emergence of macro-level dynamics and properties. Those dynamics and properties are defined as ‘emergent’ because they cannot be traced back to the behavior of the system’s individual constituents (Mitchell, 2011; Thurner et al., 2018). A glass of water, for instance, cannot be understood by simply looking at isolated molecules of hydrogen and oxygen, but needs additional information about the ratio of molecules and environmental parameters like pressure and temperature. Similarly, in social systems, a conflict cannot be traced back to the position of single individuals but is the product of their interactions and relative positions. Moreover, those dynamics may be nonlinear, that is, a small change at the micro-level may have large effect at the macro-level and vice-versa. At sea level, for instance, warming up water from 10° to 80° Celsius does not change its chemical state, while going from 99° to 100° Celsius leads to a phase shift from liquid to gas. Similarly, in social systems, conflicts or opinion dynamics are nonlinearities due to negative or positive feedback loops that absorb or amplify information cascades and can lead to conflict outbreak (Lee, Daniels, Myers, Krakauer, & Flack, 2020) or shifts of popular opinion (Bassett, Alderson, & Carlson, 2012; Castellano, Fortunato, & Loreto, 2009; McBurnett, 1996).

Properties of complex systems	Equivalents in policy processes
Many components	Many actors and events interact in policy processes
Heterogeneity	Policy actors have different motivations, expertise, roles, resources, etc.
Interaction in networks	Policy actors can form alliances and interact in policy arenas
Environmental constraints	Policy actors' behavior is shaped by environmental events, institutional structures and social cues
Coevolution	Policy actors update their focus and beliefs as a function of their interactions with other policy actors
Emergence	Agendas and policies are macro-level outputs that cannot directly be linked to micro-level configurations
Nonlinear dynamics	Opinion dynamics or the behavior of public budgets are results of positive and negative feedback loops

Table 1: The hallmarks of complex systems in policy processes

These properties have been widely used in policy process theories. First, a group of scholars has explored the application of complexity theory to policy processes (Cairney, 2012; Geyer & Rihani, 2010; Klijn, 2008; Morçöl, 2013; Teisman & Klijn, 2008). They define policymaking processes as networks of policy actors who have different personal characteristics and participate in an informal and formal process of collective decision-making. This process depicts erratic features which they link to the ideas of emergence and nonlinearity. The emergent properties are either coalitions, agendas, policies, or decisions. Second, there is an implicit convergence among the theories of the policy process as reported in the work of Sabatier & Weible (2007; 2018). The following are non-exhaustive examples of this convergence.

The advocacy coalition framework proposes a framework to define the co-evolution of coalitions as a function of their interactions and the resulting adaption of individual's beliefs (Pierce, Giordono, Peterson, & Hicks, 2019). The multiple streams framework characterises policymaking as consisting of different flows of activity that, when coupled, can lead to drastic, sudden changes at the system level. This coupling is often due to the behavior of a small group of actors and to changes in policy environments (M. D. Jones et al., 2016). The policy feedback theory conceptualises feedback loops from policies to policy processes, thus theorising amplification or absorption dynamics between the products of policies processes and their inner workings (Mettler & Sorelle, 2014). Policy learning defines the network-based dynamics of continuous adaptation of policy actors as a function of new information, thus portraying the adaptive capacity of the participants of policy processes (Moyson, Scholten, & Weible, 2017). Last but not least, punctuated equilibrium theory empirically identifies power-laws in budget changes and theorises mechanisms of inertia and sudden change based on positive and negative feedback loops within micro-level processes (Baumgartner, Jones, & Mortensen, 2018).

All of these theories converge on the view that policymaking consists of bounded-rational, attention-limited (thus adaptive) actors whose interactions lead to system-level outcomes such as policies. The presented theories have been applied to numerous case-studies and seem applicable in democratic and authoritarian political contexts alike (Chan & Zhao, 2016). Comparative studies in punctuated equilibrium theory, for instance, have found empirical regularities across almost all democracies as well as China and international organizations (B. D. Jones et al., 2009; Lam & Chan, 2015; Lundgren, Squatrito, & Tallberg, 2018). Table 1 summarises the properties of complex systems and their equivalents in policy processes. Overall, the literature has progressed on understanding both micro- and macro-level dynamics (Moynihan, 2018).

All in all, scholars of policy processes have been increasingly conceptualizing policy processes in a way that exhibits the hallmarks of complex systems (Geyer & Cairney, 2015). Whether they will explicitly use complexity theory as their handle is a question related to disciplinary tradition and attachment to the particularities of certain theoretical frameworks. As for this paper, the diagnosis is clear: the analysis of policymaking processes would highly benefit from methodological advances in complexity science. As it is already used at least implicitly, all it may require is a methodological shift, which we address in the next section.

1.2 Methodological frontiers in examining complexity

While the application of complexity theory - even if implicit - to understand policy processes is increasingly accepted, the employed methods fall short of providing system explanations that are in line with complexity science. In this section, we present the main methodological approaches used in the current policy process literature, discuss their limitations and outline recent methodological progress that suggests a promising but rocky path for a computational turn of policy process studies. Table 2 summarizes the key references that contribute to the study of policymaking as a complex system.

The works that explicitly use complexity theory to describe policy processes is mostly theoretical. They focus on refuting or validating the use of complexity theory in policy process studies and make recommendations for future scholarship, such as the unification of theories, moving away from simplistic approaches, or recognising unpredictability (Geyer & Cairney, 2015). In contrast, the scholarship applying the theories implicitly is either purely theoretical, purely qualitative or purely quantitative.

The different theories are subject to debate. Scholars unpack criticisms and refine theoretical mechanisms, often as a result of empirical insights and key events that shed light on policy processes (e.g. 2008 economic crisis, Fukushima's tsunami or Brexit (Giger & Klüver, 2012)).

When used as lenses to understand specific cases, those theories are usually accompanied by qualitative methods, ranging from ethnographic research, to interviews and social network analyses (Keman & Woldendorp, 2016). Those approaches often remedy the lack of granular data on the micro-level dynamics of policymaking processes. Therefore, scholars need to conduct (semi-)structured interviews with policy actors to access information about beliefs, behaviors, network structures and power dynamics. These research results are often used for thematic analyses or critical analyses that deconstruct social realities (Levy, 2007).

Scholars also employ quantitative methods to validate or refute theories. Surveys, Open Government Data, media data, or the Comparative Agendas Project (Baumgartner, Breunig, & Grossman, 2019) provide sources of micro-level and macro-level data on policy processes, allowing the analysis of the relationship between changes at the micro-level and their impact on changes at the macro-level. The methods employed are either descriptive statistics or statistical models such as linear regressions (Curini & Franzese, 2020).

The current methods - theoretical, qualitative or quantitative - constrain the potential of policy process studies unless complemented. First, the literature that ties together complexity theory and policy processes mostly focuses on the meaning of complexity as a concept applied to policymaking. It is about building a narrative and a set of mental models to understand complex policymaking systems. While we think building this narrative is important to change paradigms in social science, it often falls short of identifying empirical facts and laws of policymaking systems. Therefore, it is insufficient to describe empirical insights about the complexity of policymaking systems.

Second, building a complexity narrative of policy processes also opens the door to relativism. Especially when associated with concepts from chaos theory, the complexity narrative attracts proponents of postmodernism, interpretivism and more (Morcol, 2001). While these critical schools of thought have led to important reflections about the social sciences and the role of the limitations and power of researchers in attempting to understand social phenomena, they tend to suggest that knowledge exists only if highly nuanced (Healy, 2017). This assumption, often implicit, has fostered doubts about the value of abstraction, formalization, quantitative and computational methods

(Drucker, 2016), which has slowed down the adoption of computational methods in policy process studies.

Third, policy process studies are divided into roughly two clusters of scholars. On the one hand, scholars push very reductionist approaches and focus on e.g. linear relationships between micro-level and macro-level data through black-box, statistical models. On the other hand, scholars embrace all-encompassing approaches and resort to long essays or text-books to describe complexity. These two groups seem to position themselves differently with respect to modeling social phenomena. The first group focuses on models that do not account for the characteristics of complex systems while, the second group, does not seem to engage with any form of formal modeling. The computational turn is a way to reconcile these clusters, by putting together scientists who formalize policymaking systems and others who have developed detailed theories thereof. By facilitating this turn, we want to motivate the search for more complex but humble models that do not try to hide their assumptions and simplifications but try to test specific theoretical propositions as best as possible.

However, it is important to highlight relevant quantitative advances in policy process studies. First, statistical analyses applied to punctuated equilibrium theory using leptokurtic distributions have been widely tested and generated empirical regularities across countries (Chan & Zhao, 2016; Sebók & Berki, 2017; Sharp, 2019). They notably show a power-law distribution of changes in public budgets. These robust findings are one of the first results that look like the ones complexity science generates on other systems: macro-level patterns that are found across systems, which suggest that systems behave the same way and thus lead to the formulation of hypotheses for micro-level dynamics. In fact, scholars have found similar findings in conflict dynamics, where the frequency and severity of conflict also follow a robust a power-law (Clauset, 2018; Richardson, 1960).

The second advancement refers to the development of network or agent-based models to formalize the micro-level dynamics of policy processes and generate macro-level phenomena (Cioffi-Revilla & Rouleau, 2010; Klein, 2017; Kollman & Page, 2006; Richards & Doyle, 2000; Thomas, 2017), following what complexity scientists have been applying to other areas of social science (Axelrod, 1997; Bhavnani, Donnay, Miodownik, Mor, & Helbing, 2014; Guerrero & Axtell, 2013). This progression is promising as, so far, quantitative models were statistical (i.e. modeling of data) while in network and agent-based models, the idea is to model not just observations but the generating mechanisms of systems themselves. A third advancement consists of deep learning models using decision trees and forest fire dynamics to replicate power-laws in budget changes and link them with micro-level processes (Hegelich, 2016, 2017). While in their infancy, these models – network, agent-based and deep learning – are increasingly connected to policy process theory, and thus illustrate the computational turn that goes hand in hand with parallel scholarship. However, to date, these models have not generated particularly useful insights beyond unpacking the barriers to computational modeling of policy processes. They are also published in journals poorly known to policy process scholars. In the rest of this paper, we propose avenues to further progress.

2 Computational models to understand policymaking

The advent of computers was essential for the study of complex systems. They enable the formal and algorithmic study of systems in a way that can be anchored in and validated by empirical data. Computers are a powerful tool to study systems whose behaviors are too complex for human intuition to grasp and cannot be reduced to linear relationships. Computational methods include any computational tool that serves to collect and model data as well as model systems (Lazer et al., 2020). In this paper, we focus on computational models of systems and this section briefly

Reference title	Author(s) & year	Implicit or explicit use of complex system	Type of contribution	Contribution	Formal modeling
Complexity in Public Management	Teisman and Klijn (2008)	Explicit	Theory	Mapping of properties of complex systems on the characteristics of public policymaking processes	NA
Complexity and Public Policy: A New Approach to Twenty-First Century Politics	Geyer and Rihani (2010)	Explicit	Theory	Mapping of properties of complex systems on the characteristics of public policymaking processes; and discussions of implications	NA
Complexity Theory in Political Science and Public Policy	Cairney (2012)	Explicit	Theory	Discussion of the use of complexity to understand policymaking processes and implications for the field	NA
A Complexity Theory for Public Policy	Morçöl (2013)	Explicit	Theory	Mapping of properties of complex systems on the characteristics of public policymaking processes	NA
Handbook of Complexity and Public Policy	Geyer and Cairney (2015)	Explicit	Theory; review	Collection of chapters on the use of complexity theory to explain policymaking processes; and description on the use of agent-based models in policy process studies	Steps for agent-based modeling
Common Approaches for Studying advocacy: Review of Methods and Model Practices of the Advocacy Coalition Framework	Pierce et al. (2019)	Implicit	Theory; review	Description of the use of co-evolutionary coalitions to explain policymaking processes; report that 70 to 100% analyses are qualitative	NA
A River Runs Through it: A Multiple Streams Meta-review	M. D. Jones et al. (2016)	Implicit	Theory; review	Description of interplay of actor and environment to explain policymaking; report 88% of studies are qualitative	NA
Policy Feedback Theory	Mettler and Sorelle (2014)	Implicit	Theory	Describe feedback loop between policy outcome and policy processes	NA
Policy Learning and Policy Change: Theorizing their Relations from Different Perspectives	Moynon et al. (2017)	Implicit	Theory; review	Describe adaptive behaviour of individuals and groups	NA
Punctuated Equilibrium Theory: Explaining Stability and Change in Public Policymaking	Baumgartner et al. (2018)	Implicit	Theory	Describe power-law in budget changes and theorise positive and negative feedback loops at the micro-level	Statistical modeling
Political Complexity: Nonlinear Models of Politics	Richards and Doyle (2000)	Explicit	Review of models	Collection of models of politics, with unclear link to policy process theories	Various
Computational Methods and Models of Politics	Kollman and Page (2006)	Explicit	Review of models	Review of models of politics, mostly focusing on electoral systems and institutional design	Agent-based modeling
MASON RebeLand: An Agent-based Model of Politics, Environment, and Insurgency	Cioffi-Revilla and Rouleau (2010)	Explicit	Exploratory modeling	Formalization of a large-scale agent-based model with unclear link to policy process theories	Agent-based modeling
Policy Emergence: an agent-based approach	Klein (2017)	Explicit	Exploratory modeling	Formalization of an agent-based model based on combination of policy process theories	Agent-based modeling
Modeling Contagion in Policy Systems	Thomas (2017)	Explicit	Exploratory modeling	Modeling of attention contagion in policy networks, with clear link to policy process theories	Agent-based modeling
Deep Learning and Punctuated Equilibrium Theory	Heglich (2017)	Explicit	Exploratory modeling	Modeling of patterns of attention	Deep neural networks

Table 2: Sixteen key references that make contributions to studying policymaking as a complex system

presents what computational models are and why they are useful, before outlining a process of evidence-driven computational modeling for policy processes.

2.1 What are computational models?

Before the advent of personal computers in the 1990s, the formalization of social systems fell short of replicating complex dynamics. They were able to generate either deterministic dynamics (Richardson, 1960) or chaotic dynamics (Saperstein, 1984). Later on in the 1990s, networks of computers could facilitate purely theoretical computational research in e.g. international relations and the validation or refutation of traditional theories (Duffy, 1992). As of the 2000s, computational models became more precise and increasingly empirically validated (Clauset & Gleditsch, 2012; Weidmann & Salehyan, 2013). In sixty years, the field of social system formalization moved from systems of equations to computational toy models and on to empirically-validated computational models. It is this latter, empirically-validated computational type we want to encourage in policy process studies.

Computational models can be referred to as dynamic formalisms because they combine two core components. First, models are implemented as systems of equations, computer code or a mix of both (Thurner et al., 2018). This process of formalization aims to reduce the description of system parts and mechanisms to a minimum of components in a way that is theoretically and/or empirically valid. To model complex systems, formalizations often include update rules or feedback loops, which are iterative mechanisms that lead to adaptation, absorption or amplification. Second, computational power is used to run the formalization and activate models’ iterative processes. The iterations then generate system dynamics.

To examine social systems, a prominent class of computational models are agent-based models (Page, 2008). These models formalize the characteristics and adaptive behaviors of agents (e.g. policy actors), their mode of interactions and network structures (e.g. topology of policy networks) and the macro-level properties (coalitions, attention dynamics, collective decisions, ...) that result from iterative processes of agent-level adaptation (e.g. opinion changes, decisions, etc.). Agent-

based models are increasingly popular in political science, especially in the study of intra-state conflict (Neumann & Lorenz, 2019).

“Is it possible to model human behavior?” is a question that is repeatedly raised in political science departments and conferences. This paper builds on the insight that, in fact, all descriptions of human behavior are models. Verbal descriptions may seem more faithful because of their fuzziness and room for interpretation, allowing people to project their own models on what is meant. Yet, current verbal descriptions of policy processes based on complexity theory are difficult to test empirically because they are often too general. Therefore, a minimum level of formalization is needed. Similarly, mathematical models without implementation fall short of eliciting complex dynamics. It is the coupling of mathematical formalization and computational power that can usefully model human behavior and explain social dynamics (Helbing et al., 2015). All models are wrong, some models are useful (Epstein, 2008). Any modeling effort needs to state its limitations clearly, but models do not need to be perfect. They need to fit the theory and the data and be tested against reality.

2.2 The value of computational models

Computational models are particularly useful because they formalize and generate dynamics of iterative, stateful, path-dependent processes. This is important because social systems feature plenty of iterative processes. Conflicts, opinion dynamics or economic supply and demand relationships, for example, are all iterative processes that boil down to how agents update as a function of the behavior of other agents. Modeling such iterative processes enabled the study of emergent cooperation among heterogeneous agents, for instance (Axelrod, 1997).

Computational models also enable the study of the dynamics of systems of equations that are not analytically solvable. Differential equations of complex systems rapidly become too complex to be solved by hand and most are non-deterministic (Thurner et al., 2018). Therefore, implementing those equations as computational models enables researchers to generate dynamics and then analyse them statistically. Relatedly, while it is extremely difficult to make point-predictions about complex systems, computational models can make accurate predictions of outcome distributions by running thousands of simulations (Lazer et al., 2020; Thurner et al., 2018).

Computational models enable to generate data on either micro-level or macro-level dynamics when no real data is available (Epstein, 2006). Therefore, models may be built on theory and then used to generate data with which statistical analyses are possible. This is particularly useful to examine complex systems for which data collection is very difficult (secret processes, or dangerous locations) and to analyse counterfactuals in possible future outcomes, assuming the model has been sufficiently validated (Page, 2018).

Computational models also allow researchers to reconcile the study of micro-level and macro-level dynamics. As the mainstream literature tends to dissociate both (e.g. micro- versus macroeconomics, or policy analysis versus behavioral public policy), the use of computational models can link macro-level dynamics to the behavior of their constituent parts. They directly integrate the concept of emergence and can produce the nonlinear dynamics associated with emergent phenomena.

In social systems, there is often one history that cannot be repeated, which prevents the rigorous understanding of what-if scenarios. Computational models allow researchers to repeat artificial scenarios and compare them to each other as statistical equivalents (Thurner et al., 2018).

It is, however, very challenging to reliably attain the added-value we proposed. The next section presents a guide to maximise the success of computational modeling efforts.

2.3 From model conceptualization to the identification of signatures of policymaking systems

Recent literature on computational social science has aggregated pieces of advice for creating valuable models. Bhavnani and colleagues (2020) summarise this advice in a process to develop robust evidence-driven computational models. We adapt their process to policy process studies. The advantage of this process is that it places computational modeling within a larger scientific process, thus linking different stages of research design.

Figure 1 illustrates the resulting five step process: (1) model conceptualization; (2) model implementation; (3) data construction; (4) model validation and refinement; and (5) counterfactual analysis. As such, it guides researchers in the identification of the signatures of complex policymaking systems. While the figure depicts a linear process, it is inherently recursive. In the rest of this section, we describe each step and link it to the policy process literature.

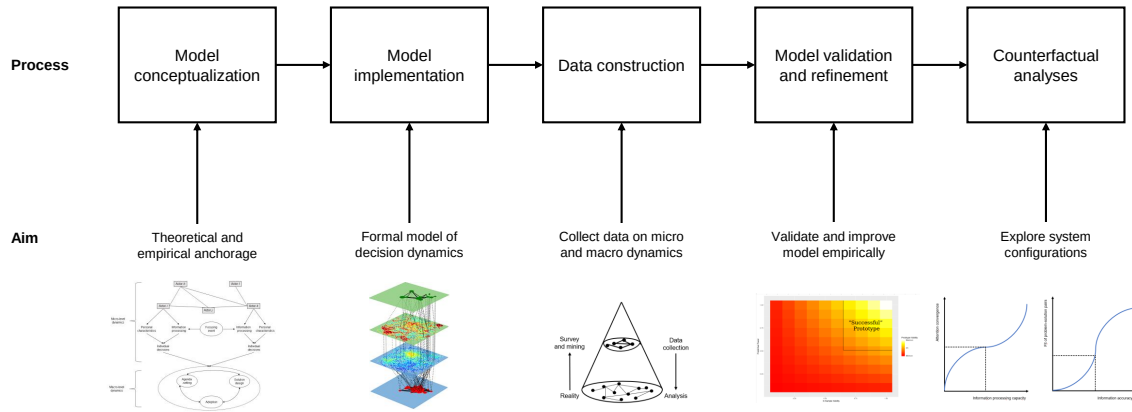


Figure 1: A process for computational modeling in policy process studies (adapted from Bhavnani et al., 2020)

Model conceptualization: the first step of computational modeling is the definition of parts, mechanisms and outcomes. Model conceptualization is, in a way, a synthesis of many different information sources. To conceptualize a model of policymaking, it is possible to draw from existing theories of policy processes (Weible & Sabatier, 2018), existing case-studies, expert opinion, interviews and surveys. Altogether, these sources of information enable the specification of system parts, the selection of specific mechanisms and link micro-level with macro-level dynamics. A crude, conceptual model of policymaking, for instance, can be deduced from punctuated equilibrium theory as shown in Box 1.

Box 1. A simple verbal model of policy processes based on punctuated equilibrium theory

Policy actors have two adaptive characteristics which is their current attention and opinion, they interact within a decentralized network structure, and update their attention and opinion

as a function of their neighbours’ characteristics. Actors pursue two strategies: trying to reach better positions in the network and forming coalition with value-aligned actors. Iteratively, this process leads to aggregate patterns of attention such as periods of inertia and periods of change, and contagion of opinion. (B. D. Jones & Baumgartner, 2005a, 2005b).

Model implementation: once the model is conceptualised and matches descriptions and data from available sources of information, one can implement it. Implementation includes formalization of the model as systems of equation and computer code. The formalization specifies the relationship between variables and their associated probabilities, how the model aggregates observables (like patterns of attention), and how it reports results. Formalization often is a crucial step to test the robustness of the conceptual model because it forces researchers to spell out modeling assumptions and will thus challenge fuzzy definitions. Relevant mathematical and software programming tools must be employed and their use justified (see next section). Model implementation also involves many checks to test whether the implementation produces the intended dynamics. Sensitivity analyses are useful to check whether the model behaves in a realistic manner. In policy process studies, the implementation of the conceptual model could draw from differential equations that help formalize co-evolving networks (Thurner et al., 2018). See the next section for an approach to formalize policy processes.

Data construction: once the model is implemented, it is called a toy model, as its results will not be empirically valid and can only serve exploratory purposes. To produce empirically valid results, the third step is to identify, collect and prepare data to seed and validate the model (see next paragraph). Data is required to specify the initial conditions of the model, such as the number of policy actors, their initial focus, the network topology and the frequency and size of exogenous signals, such as media coverage. To validate the simulated results - such as aggregate patterns of attention, the rate of new decisions, and the evolution of network topologies - more data is required.

Finding useful data is a challenge. Data on micro-level processes is scarce and often qualitative. There is some progress in collecting such data under the label ‘big data’ by relying on email data to identify network topology and social media data to identify attention patterns and political behavior (Donnay, 2017). Surveys and behavioral experiments can also provide complementary information, especially on behavioral profiles (Moser & Kalton, 2017). Data on macro-level processes, such as agendas, attention or decisions, is easier to find in a quantitative format. The Comparative Agendas Project centralises data on public budgets (Baumgartner et al., 2019) and the Open Government Data group reports decisions and bills issued by national governments (Ubaldi, 2013). In most cases, the identified data will not be directly usable to seed and validate the model. It will require “construction” of datasets that can be imported into the model by, for instance, matching dataset variables with model parameters. Or by drawing probability distributions from datasets to define update rules, frequency of exogenous signals, etc.

Model validation and refinement: once data is prepared, researchers can use it to seed a model’s initial conditions and validate simulated outcomes. Multiple methods can be used to measure the fit of simulated results with empirical data, including the root-mean-square-deviation, but also precision, recall and F1 scores. F1 scores are particularly useful as they provide information on how many observations the model gets right or wrong, instead of a simple overall correlation or deviation (Goutte & Gaussier, 2005). Data can be used to test internal validity (using in-sample data from a specific case) and then external validity (using out-of-sample data from other cases). Here again, the insights from punctuated equilibrium theory are useful. The power-law of public budgets has been empirically validated across contexts. It would be interesting to build a model of the micro-level

dynamics that lead to its emergence and then explore whether its mechanisms are valid in various contexts.

Counterfactual analysis: once models are empirically validated, at least internally, it is possible to use them as artificial laboratories to explore their signatures. Conducting counterfactual analyses allows to understand how the system behaves when different initial conditions are chosen. For instance, one can explore different levels of network centrality, different attention probability updates, different frequencies and sizes of exogenous signals, and explore how the system behaves as a function of those changes. In complexity science, it is essential to explore critical points - the equivalent of the 99 to 100 degree shift in water temperature that leads to a phase transition. For instance, does increasing network density from one specific value to the next lead to radical changes at the macro-level? Those analyses could allow the robust identification of the drivers of policymaking. As such, they can then be used to generate recommendations on how to design policymaking processes (see section 4).

This five-step process allows researchers to move from the conceptual models, often described in policy process studies, to the exploration of common signatures of complex policymaking systems. It is important to iterate on these five steps, instead of trying to perfect each of them only once. The questions in the following two sections relate to modeling particular phenomena in policy process studies and avenues to be explored with this process.

3 The contours of a computational approach to model policy processes

Manoeuvring from model conceptualization to model implementation provides two key challenges. Firstly, it raises the question of finding an accurate formalization and, secondly, it requires adequate tools to implement the dynamics of said formalization.

This section reviews successful modeling approaches and sketches the contours of their implementation. Subsequently, we apply our approach to policy processes outlined in Box 1. While this section is far from exhaustive, it aims to exemplify the transition from the conceptualization to the implementation of modeling policy processes. By specifying the strategy of our approach we illustrate potential to better understand policy processes.

3.1 A case study in opinion dynamics

Our goal in this section is to retrace what makes for a successful formal model of social dynamics. We aim to highlight advantages and challenges on the path to attain a causal model and with it a mechanistic explanation of policy processes.

It is in principle hard to make a precise verbal explanation such that it uniquely defines a system in its micro- and macro-states. Or even to prove a causal relationship between them. The immediate payoff of any formal model of the policy process is a proof-of-concept for the internal consistency of stated axioms. Enabling us to explore scenarios and generate hypotheses. A prominent example being Axelrod’s computer tournaments to find optimal strategies for the iterated prisoner’s dilemma (Axelrod & Hamilton, 1981). After finding the winning strategy (“tit-for-tat”), Axelrod et al. could provide an explanation of why such behavior is common in any species with similar selection

pressure. However, the process of policymaking constitutes an open system, meaning that it is highly interconnected on many different levels of abstraction with some larger system, outside the scope of any modeling attempt. Consequently, models of such systems will most likely be unsuitable for forecasting and restricted to improve the qualitative understanding of each constituent’s contribution to the overall dynamics.

For example, R. Hegselman and U. Krause followed these considerations when developing their model of opinion dynamics (Hegselmann, Krause, et al., 2002). The goal of their model was specified as providing mechanistic explanations for the process of opinion fragmentation. What are the conditions for consensus or polarization to emerge? The first step was to formulate axiomatic rules that they believed to be the essential rules of the process. The second step was to define a corresponding formal mathematical framework that, besides allowing for further extensions, most importantly answered the following question: To what extent do the system dynamics follow from stated assumptions? In what follows, we retrace their approach and formalization and, thus, illustrate the process of formalizing social phenomena.

3.1.1 The minimal model

With n as the number of agents contained in the considered population, some discrete-time $T = 1, 2, \dots$ and some continuous opinion $x_i(t)$ with $t \in T$ for each agent i on the interval $[0, 1]$ the fundamental building-blocks for an opinion dynamics model are defined. We then want to define the most basic updating mechanism for each actor, namely taking the average over the opinion of neighbouring agents and the agent’s own opinion. However, actors do not weigh everyone’s opinion equally. Therefore we define weights for the all connections in the undirected network, resulting in the adjacency matrix $A := (a_{ij})$ of the network. As such, the updating mechanism can be compactly written as:

$$\mathbf{x}(t+1) = \mathbf{A}(t, \mathbf{x}(t)) \cdot \mathbf{x}(t) \quad \text{for } t \in T \quad (1)$$

Formalizing the general research question to: how does a given initial opinion profile of the entire network influence the final opinion distribution? Equipped with this formalization we can now investigate the most basic case with $\mathbf{A}$ as a stochastic matrix. This allows estimating convergence time to the steady-state depending on the structure of $\mathbf{A}$. We identify the network structure and opinion updating rule of each actor as the key elements that dictate the dynamics. Hence, making them the focus of further analysis (Hegselmann et al., 2002).

3.1.2 Extensions of the minimal model

Building on this minimal model, Hegselman and Krause introduced many interesting extensions, such as the property of susceptibility for each actor or “hardening of positions” where actors weigh their own opinion more strongly over time. Moreover, they analyzed non-linear extensions, which were introduced by defining conditions under which weights can become zero - meaning actors are ignoring each other. One possible implementation is to define a confidence level ε_i to each agent similar to the Deffuant-Weisbuch model (Deffuant, Neau, Amblard, & Weisbuch, 2000). Such models are called bounded confidence models, where the confidence level acts as threshold parameter that defines the level of discrepancy of opinions beyond which actors ignore each other. Further research investigated the implications of asymmetric confidence intervals between the agents and of the

underlying network structure. From this we learn that choosing the right level of abstraction (i.e. minimal model) is essential and allow for useful extensions at a later point. Ideally, we do not arrive at a single, large and specific self-consistent model, but rather at a cluster of different models exploring implications of different assumptions.

3.2 A simple model of policy processes

Drawing from the outlined considerations, we now propose a minimal model for policy processes based on the verbal model describe in Box 1.. We first make our assumptions for the system explicit and subsequently propose a formalization of them. There are three components to our model: (1) the internal state of actors, (2) the network topology and (3) the updating mechanism.

3.2.1 Components of the model

(1) The internal states are characterized by the actors attention and their deeper beliefs/opinions. While both change over time, we assume that they happen at different timescales, with attention changing in much shorter intervals (B. D. Jones & Baumgartner, 2005a). Additionally, we believe that so-called “focusing events” shape attention dynamics but are negligible in opinion dynamics. Furthermore, we incorporate bounded rationality by limiting the attention that can be distributed among different topics for each actor. Lastly we equip every actor with an idiosyncratic strategy preference, which will be specified in (3). This results in the following: Actor i will have an opinion vector $\mathbf{x}_i^0(t) \in [0, 1]^n$ and a corresponding attention vector $\mathbf{x}_i^A(t) \in [0, 1]^n$ that, together with the strategy $s_i \in [0, 1]$ will define the internal state vector $\mathbf{X}_i(t) \in [0, 1]^{2n+1}$.

(2) A node in the network can have an unlimited number of edges. However, we take limited processing capabilities into account by limiting the sum over all weights for all actors to a constant, similar to the concept of Dunbar’s number (Dunbar et al., 2012). The strength of the connection between two actors is then some function of the edge weight connecting them, alignment of their internal states and common connections. The network is adaptive in nature, re-weighting and rewiring its edges over time. This corresponds to fluctuations in the intensity and composition of their relationships. We assume that the re-weighting process is more essential and therefore neglect the rewiring process. Additionally rewiring often increases the computational cost for simulations immensely. However, we will investigate the effect of introducing new and removing old nodes into the network. As a consequence we have some stochastic matrix $\mathbf{A}(t) \in R^{n \times n}$ that defines the network.

(3) Lastly the updating mechanism constitutes the core of our model of policymaking systems. We interpret the underlying social networks as multi-layered co-evolving networks (Thurner et al., 2018, p. 21-25). Similar to the model by Hegselman and Krause (2002), the evolution of every actor’s state and the network is some function $\mathcal{F}$ of its previous internal state and the weight matrix:

$$\{\mathbf{A}(t+1, \mathbf{x}(t+1)), \quad \mathbf{x}(t+1)\} = \mathcal{F}[\mathbf{A}(t, \mathbf{x}(t)), \mathbf{x}(t)] \quad \text{for } t \in T \quad (2)$$

This mutual dependence of opinion dynamics and network topology accounts for the co-evolutionary properties of our model. Subsequently, we will postulate further assumptions on the function $\mathcal{F}$ that are valid for policy processes in general.

Each actor has to update his internal state, namely his attention and opinions. Furthermore, every actor is bounded-rational and has imperfect memory or learning capabilities. We can encapsulate these dynamics in a “stress” function $\mathcal{H}$ each actor tries to minimize:

$$\min\{\mathcal{H}(\mathbf{A}(t), \mathbf{x}(t))\} \quad (3)$$

This approach closely resembles other successful model of opinion dynamics based on the Ising model (Galam, 2008). $\mathcal{H}$ should take the states of the actors local/global neighbourhood into account. Additionally, this function will have a stochastic/perturbative component that accounts for the previously mentioned cognitive limitation. In the limit case of no cognitive ability, the actions should appear to be random. The weights of the edges account for the influence that each neighbor has on one particular actor. Optionally, this influence can be dependent on more parameters than weights in $\mathbf{A}(t)$. The update of each actor’s opinion therefore depends on the network’s topology.

Furthermore, actors will have two strategies that guide their interactions. First they are driven to increase their degree centrality (i.e. importance in the network) and second they want to form coalitions with value- and goal-aligned actors. We denote the function of the gained utility of these strategies with C_e for a centrality-driven and C_o for a coalition-driven strategy respectively. These two strategies specify the preferred attachment-style of a certain agent (i.e. the connections they want to establish or maintain). Therefore they constitute the driving mechanism of the evolution of the topology. Each actor’s idiosyncratic strategy is determined by the previously defined constant s_i in the following way:

$$\max\{s_i C_e(\mathbf{A}(t), \mathbf{x}_i(t)) + (1 - s_i) C_o(\mathbf{A}(t), \mathbf{x}_i(t))\} \quad (4)$$

Note that this optimization only takes existing edges into account and reweighs them such that the term is maximized for a certain strategy. The previously undefined function $\mathcal{F}$ can now be identified as the application of these two optimization problems for a given actor i in the system.

3.2.2 Procedure and Extensions

We have outlined the fundamental mechanism of the model. Formulating these functions more explicitly will need empirical justification. Nevertheless, we can determine next steps of analysis. After implementing the model we will investigate its properties through established schemes for sensitivity analysis and Monte Carlo methods, answering questions such as: How does the system’s steady state depend on its initial configuration? How sensitive is it to the actors rationality/randomness? How does its rate of convergence towards the steady state depend on factors such as the dimensionality of opinions or the capacity for attention? Do we already observe punctured equilibria and how can they be explained without external shocks? Do certain strategy profiles correspond to specific known groups such as “policy entrepreneurs” (Aviram, Cohen, & Beeri, 2020)? If so can we validate their suspected influence on the system?

Additionally we want to make two implicit assumptions of our approach explicit: First, the proposed model is time-discrete. Switching from discrete to continuous time models often changes the dynamics profoundly. Consequently, we need to observe closely if this may distort the system behavior we observe. We have chosen discrete time due to its computational advantages, but it is not necessary for analysis in principle. Second, the model’s time evolution only depends on the state of the previous time step, making it markovian. This means it does not remember its history. In many complex systems this is an acceptable approximation and yields models with large explanatory

power regardless. Transferring this approach to a political system is not trivial and requires careful consideration.

Exploring the proposed model would already yield insights on the behavior of policy processes. In particular, it would provide a way to explore their dynamics in a way that reconciles both micro- and macro-level dynamics. Such efforts can be expanded upon by adapting the model to allow for exchanging actors over time, external events that have a global effect on attention (i.e. focusing events). Once we have established an understanding of key properties, we can go on to introduce more refined rewiring processes, more complex events and more heterogeneous properties for the actors.

4 A research agenda for computational policy process studies

Taking stock, we delineate two main research avenues to achieve the computational turn of policy process studies. First, we outline areas where progress is to be made - from formalizing policy processes to generating empirical evidence about the drivers of policy change. Second, we articulate three contributions that computational policy process studies can make in the near future.

First, the computational modeling process sketched in section 2.3. highlights the necessary steps to move from a conceptual understanding of policy processes to the identification of their empirical signatures. We see five areas of progress. First, the computational approach outlined in section 3. must be expanded and refined. It is important to carefully select which concept of policy process studies deserve formalization and then identify promising tools. Second, once the approach provides isolated formal components of policy processes, one can select and assemble them to design toy models. Toy models will be used to elicit dynamics that can be validated theoretically. The idea is to check whether results come close to the expectations in the literature. Third, more effort must be put into collecting data on the characteristics, behavior and network connection of policy actors. Building datasets of micro-level determinants is crucial to advance empirical research and examine how micro-level dynamics lead to macro-level outcomes. Fourth, toy models can be expanded and seeded and validated with empirical data. Fifth, empirically-validated models can then be used to explore counterfactual scenarios and identify critical points of policymaking systems.

These five steps can apply to single projects but also to the entire field of computational policy process studies. As such, as computational projects multiply, they can feed into each other and contribute to a better understanding of how to model policy processes, increased data validity, and the identification of key critical points.

Second, if the proposed process is implemented and benefits from a strong community that contributes to its development, we expect the field to progress towards three core advances. First, as things stand, the literature on policy processes has identified a robust power-law of public budgets, which is a decent proxy for policy change (B. D. Jones et al., 2009). The same literature has identified drivers of political behavior and policy networks (Moynihan, 2018). Yet, both streams are somewhat detached from one another (Thomas, 2017). Ontologically akin to Schelling’s work on Micromotives and Macrobehaviors (Schelling, 2006), we expect the computational turn to reconcile micro and macro level and provide empirical explanations of how policies change as a function of small changes at the micro-level. While public budgets depict empirical regularity across contexts (Baumgartner et al., 2018), the driving mechanisms may be contextually different (Chan & Zhao, 2016; Sharp, 2019). Moreover, the formalization of complex policymaking systems will also enable the exploration of a large set of correlates and test the sensitivity of initial conditions (e.g. network

structures), the formation of coalitions and the effects of exogenous perturbations. As such, this avenue looks very similar to the computational developments in the study of conflict. A power-law of the frequency and severity of conflict was discovered in the middle of the twentieth century and has been confirmed over time (Clauset, 2018; Richardson, 1960). In parallel, other scholars have theorized mechanisms of inter-state and intra-state conflicts (Kalyvas, 2006). Only very recently, empirically valid computational models formalize micro-level dynamics and generate validated power-laws of conflicts (De Mesquita, 2006; Lee et al., 2020).

Second, reconciling micro-level dynamics and macro-level outcomes computationally and empirically can, in turn, allow the exploration of levers (Meadows, 1997) for policy change. For instance, how does shifting information provision (e.g. increasing information quantity in the system) compare to shifting information processing (e.g. allowing actors to process more or less information) versus changing network structures (e.g. clustering or declustering networks) versus changing rules (e.g. incentives and reward structures) (B. D. Jones & Baumgartner, 2005a; van Veen, Kudesia, & Heinemann, 2020)?

Third, the examination of those different levers within formal models can shed light on how to intervene in policymaking systems. Therefore, the ultimate contribution of computational policy process studies could be to generate evidence-based recommendations to reform policymaking such that it better accounts for the bounded rationality of policy actors, the collective nature of its processes, its changing and information-rich environments and its inherent ambiguity. For example, which mechanisms can install the right feedback loops to foster an adequate division of power and better participation in decision-making?

The proposed research avenues characterise the computational turn of policy process studies. However, this turn does not imply the dominance of mathematicians and physicists within the social sciences. Historians, philosophers and political scientists do need to contribute their knowledge. The idea is to combine their questions and insights with the methods and ways of thinking from the computational sciences. It is a truly interdisciplinary project aiming to better understand the dynamics of collective action. As such, it is itself a collective endeavour that must aggregate the insights of various perspectives. Computational policy process scholars will have to drink their own medicine.

Conclusion

The 21st century is characterised by many societal challenges that require global coordination. The most impactful form of such explicit coordination is found in public policymaking. Understanding policymaking systems is thus crucial to achieving action across societal strata. Policy process studies has attempted to provide such explanations, ranging from the empirical law of budgets to the drivers of political behavior. In particular, it has converged on an understanding of policymaking systems that satisfies the hallmarks of complex systems. While the basic tenets of complexity theory are widely used in policy process studies, the use of its methodological counterparts - computational models - is rare and immature.

This methodological paper contributes to the computational turn of policy process studies. First, we surveyed the literature on complexity and policy processes and identified the current methodological frontiers in addressing complexity. Then, we provided definitions, added-values and a procedure for computational modeling. In particular, we addressed why it lends itself well to the study of policy processes. Third, we sketched the contours of a computational approach to model policymaking

systems, by exemplifying the formalization of key features of policy processes, including social networks and adaptive behavior. Fourth, we presented two research avenues for computational policy process studies: a research process going from formalization to the identification of the signatures of complex policymaking systems; and three contributions that we expect computational policy process studies to progress towards - empirical validation, identification of leverage points and mechanism design.

All in all, this paper aims to appeal to both the social scientists that want to reform the field of policy process studies and to the computational scientists who are attracted to the examination of social phenomena. By combining theories of policy processes with recent advances in computational modeling, this paper combines two fields that could contribute to a better understanding of collective action.

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